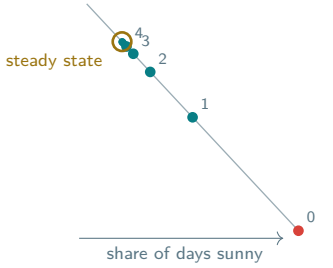
LINEAR ALGEBRA LADDER 8 OF 9

Diagonalisation and powers

Writing $A = PDP^{-1}$ turns repeated multiplication into raising the eigenvalues to a power.

1 / THE RULE AND THE PICTURE

In the eigenvector basis A is diagonal, so $A^n = PD^nP^{-1}$



a weather chain settling to its steady state

If A has n independent eigenvectors, put them in the columns of P and the eigenvalues in $D = \operatorname{diag}(\lambda_i)$. Then $AP = PD$, so

$$\boxed{A = PDP^{-1}, \qquad A^n = P\,D^n\,P^{-1}}$$

because the $P^{-1}P$ factors cancel in the middle of the product, and D^n just raises each diagonal entry to the nth power.

2 / A WORKED EXAMPLE / $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

With eigenvalues $3, 1$ and eigenvectors $(1,1), (1,-1)$ the powers have a closed form

Taking $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $D = \operatorname{diag}(3,1)$ and $P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$,

$$A^n = \frac{1}{2}\begin{pmatrix} 3^n + 1 & 3^n - 1 \\ 3^n - 1 & 3^n + 1 \end{pmatrix}.$$

For $n = 2$ this is $\frac{1}{2}\begin{pmatrix} 10 & 8 \\ 8 & 10 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} = A^2$, and for $n = 0$ it is the identity, so one formula covers every power.

3 / FIBONACCI / POWERS OF ONE MATRIX

The Fibonacci matrix Q has eigenvalues φ and $-1/\varphi$

For $Q = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ the characteristic equation is $\lambda^2 - \lambda - 1 = 0$, with roots $\varphi = 1.618$ and $\psi = -0.618$, and diagonalising gives

$$Q^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}, \qquad F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

For $n = 10$ the entries are $89, 55, 55, 34$, and the ratio of neighbours tends to the larger eigenvalue φ, since $\psi^n \to 0$.

4 / A MARKOV CHAIN / STEADY STATES

A column of probabilities that sums to 1 always has the eigenvalue 1

Each day is sunny after a sunny day with probability 0.9 and after a rainy day with probability 0.5, so $M = \begin{pmatrix} 0.9 & 0.5 \\ 0.1 & 0.5 \end{pmatrix}$ has eigenvalues 1 and 0.4.
The eigenvector for 1 is $(5,1)/6$, so the chain settles to $\frac{5}{6}$ sunny, and starting from a sunny day the share after n days is $\frac{5}{6} + \frac{1}{6}(0.4)^n$, giving $0.9,\ 0.86,\ 0.844, \ldots$ in the picture, with the gap shrinking by the factor 0.4 each step.

5 / WHEN IT WORKS

Diagonalisation needs enough independent eigenvectors

A matrix with n different eigenvalues always has n independent eigenvectors. A repeated eigenvalue may fail, as for the shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ with only one direction, and a rotation needs complex numbers. Symmetric matrices such as the first example always diagonalise with perpendicular eigenvectors, which is the spectral theorem.

ASIDE / FASTER THAN MULTIPLYING

Computing A^{1000} by multiplying takes a thousand products, while $PD^{1000}P^{-1}$ needs only two.

LINKS TO OTHER SHEETS

The Fibonacci matrix is also in *Why i makes Fibonacci inevitable*; the previous sheet is *Eigenvalues and eigenvectors* and the last is *Least squares and orthogonal projection*.

Convention. P holds eigenvectors as columns, D the eigenvalues on the diagonal; $\varphi, \psi = \frac{1 \pm \sqrt{5}}{2}$.